



Rustom-H HALE UAV, a model of Defence Research and Development Organisation (DRDO)
(Credit: en.wikipedia.org)

heading towards private sector participation. As per 'India's Defence Market Outlook 2019-2025' report, published by Market Research, between 2015-2016 and 2018-2019, out of a total 188 contracts, 121 contracts have been signed with Indian vendors, including defence public sector undertakings, public sectors undertakings, Ordnance Factory Board and private vendors for capital procurement of defence equipment.

The next decade is likely to see an exponential growth in combat systems as well as non-platform-based programmes facilitating smart battalion. Therefore opportunity for electronics in India stems across both standalone systems (as part of platforms) as well as at sub-system level for other systems. According to experts, key factors that will influence growth are:

- Modernisation of weapon platforms
- Induction of state-of-the-art weapons by the armed forces
- Impact of indigenisation and Make in India by the government

Electronics is at the heart of all modern warfare systems and, presently, the country is at the cusp of transforming the import bill in electronics that is fast competing to surpass the oil bill, presenting an opportunity for Indian companies. Prospects of growth of Indian companies will also

be driven by defence procurement and offset policies, Make in India and other key initiatives.

Most of the recent upgradation on the military side show a huge scope for electronics, including upgrades for Jaguar, Ilyushin aircraft, MIG 29, T-90 tanks and more. In addition, new ventures of LCA Tejas, AMCA, C-295 W production with Tatas, single-engine aircraft production from SAAB (Gripen), any other chosen OEM, FICV project, missile development, new aircraft carriers, submarines, MCMV, and others, all point to an increased focus on electronics. Critical avionics, night-vision devices for the armed forces, missile electronics, electronic warfare (ground-based and airborne), software-defined radios, robotics, radars of all kinds, UAVs, electronic navigation systems, weapon sensors, air defence systems and so on are the much-publicised favourites. According to experts, key demand-generating applications are:

- Weapons control systems
- Avionics
- Digitalisation of displays

In addition to that, single-board computers (new technology), advanced PCB design technology, cables and harnesses to meet aerospace standards are also in demand.

Security is a key concern for any defence application. Therefore huge

demand exists for holistic security networks and indigenously developed security devices. Despite huge opportunities, there is a demand-versus-supply gap in the strategic electronics sector with respect to availability of indigenous electronic components, products and solutions. Most components, products and solutions are still imported. Indian companies are not able to supply as per requirement, which involves low volume, high technology and high investment.

Electronic products and solutions required in India in volume will be in the areas of:

- EMS/build-to-print/line replaceable units for Indian programmes such as LCA, LCH, drones, Kamov helicopters, etc
- Rugged displays, PDAs, laptops, etc for BMS project
- Electronics components, products and solutions used in vehicles for projects like FICV, self-propelled guns and so on

Moreover, domestic capabilities are in demand to sustain and upgrade existing systems through innovative maintenance, repair and operational (MRO) solutions. Opportunities related to defence offset will be on the rise. However, there will be de-emphasis on hardware and a disproportionately larger emphasis on embedded intelligence. This will be India's second coming in the domain of software, but in terms of high value-added smart systems and analytics. Companies in India that are ready to leverage this will grow tremendously. So, offsets will have to be defined in terms of service offerings rather than hardware, to take advantage quickly.

Tectonic shift in technology

The A&D industry around the globe will focus more on disruptive technologies and fourth industrial revolution technologies, such as artificial intelligence (AI), 3D printing, fuel cells and so on.

AI is becoming prevalent across new naval technologies—missiles, navigation and aircraft—and this trend is set to increase, with intelligence increasingly devolved from platforms and systems, and injected into weapons. Recent developments